

Roadmap - Topic 7 Energy in the Atmosphere and Ocean

Learning Objectives	Vocabulary		Questions
Lesson 1: Energy in Earth’s Atmosphere			
- energy moved through Earth’s atmosphere by radiation, conduction, and convection - energy from the sun is either absorbed or reflected by the different levels of the atmosphere - the greenhouse effect traps and absorbs gases in the atmosphere	Greenhouse effect Convection Absorb	Conduction Radiation Reflect	1. How does the atmosphere keep Earth’s surface from being too hot or too cold? 2. How do both absorption and reflection of solar radiation by the atmosphere and Earth’s surface produce the greenhouse effect?
Lesson 2: Patterns of Circulation in the Atmosphere			
- wind results from differences in air pressure from unequal heating of the atmosphere, jet streams, and ocean currents - unequal heating and Earth’s rotation affect wind and weather conditions - local winds occur over short distances and affect local weather; global winds occur over large areas and move around the globe - global convection currents and the Coriolis effect interact to make the wind pattern of calm areas and global wind belts around Earth	Wind Sea Breeze Land Breeze Anemometer Trade Winds Prevailing Westerlies	Coriolis Effect Jet Stream Doldrums Horse Latitudes	1. What does an anemometer measure? 2. Model the pattern of sea breezes and land breezes between day and night. 3. How does unequal heating of the Earth’s surface produce global winds? 4. How does Earth’s rotation cause the Coriolis effect? 5. Why are global wind patterns and climate tied so closely to latitude? 6. How does the jet stream affect local temperatures?
Lesson 3: Patterns of Circulation in the Ocean			
- global winds drive surface currents - unequal heating and the rotation of Earth combine to produce the patterns of ocean circulation - ocean currents redistribute Earth’s energy	Current El Niño La Niña	Climate	1. How do wind patterns affect ocean currents? 2. How do ocean currents create the difference in climate between Eastern and Western sides of a land mass?